Project Executable Files Documentation

Date	18 October 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

Comprehensive Guide to CrimeVision AI Executable Scripts:

Executable Script	Command Line Execution	Parameters / Flags	Outputs & Side Effects
backend/app.py	python backend/app.py	--port 5000 (configurable in config.py)	Starts Flask HTTP server; registers Blueprints; serves SPA at http://localhost:5000.
backend/video_module/opencv_detector.py	POST /api/video/opencv-predict	video_path or multipart file	Executes OpenCV frame-by-frame analysis predicting people, vehicles, objects, 2D boxes, 3D trajectories, and timeline.
backend/test_external_data.py	python backend/test_external_data.py --all	--image <path> --audio <path> --video <path> --all	Automated batch forensic evaluator; runs vision/audio/video pipeline and generates Court PDFs in /data/reports/.
backend/generate_external_dataset.py	python backend/generate_external_dataset.py	None (Autonomous)	Synthesizes 3 external images, 3 audio files, and 2 video CCTV clips in backend/data/external_samples/.
backend/generate_samples.py	python backend/generate_samples.py	None (Autonomous)	Generates default benchmark evidence (sample_crime_scene.jpg, robbery_witness.wav, surveillance_store.mp4).
backend/tests/test_validation.py	python -m unittest backend/tests/test_validation.py	-v (Verbose mode)	Executes 5 integration tests (Health, Fusion, Chat, Video Timeline, PDF Generation).
generate_documentation_pdfs.py	python generate_documentation_pdfs.py	None (Autonomous)	Compiles all official SmartBridge submission PDFs across all 7 project phases.
generate_architecture_diagram.py	python generate_architecture_diagram.py	None (Autonomous)	Renders high-resolution 1600x1000 system architecture diagram image (architecture_diagram.png).